Mathematics: analysis and approaches
Standard level
Paper 1 — markscheme
Topic 4 Statistics — box-and-whisker diagrams

5 November 2026

W4ss Academy practice set 1 | **Topic** Topic 4 Statistics — box-and-whisker diagrams | **Ref.** W4SS–MAA–QB–P1–01

Student name

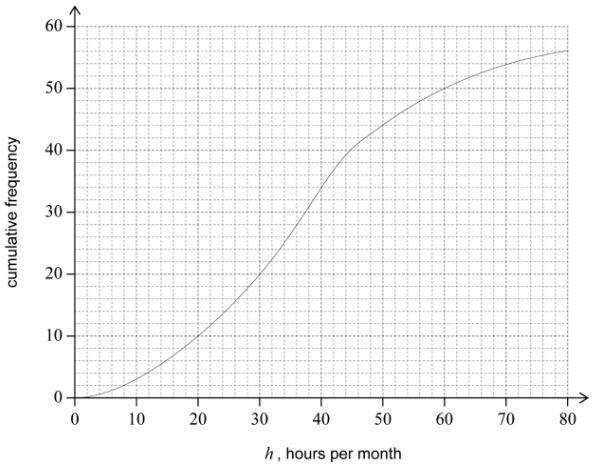
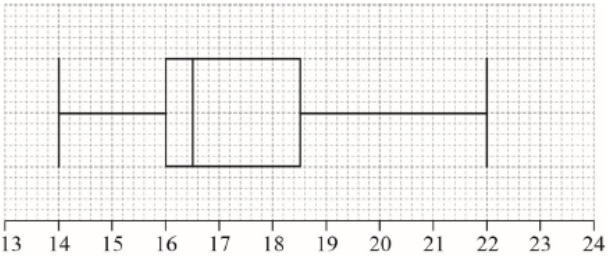
Paper 1

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[44 marks]**.

Section A

Question	Marks	Markscheme
1.	**[2]**	Cafeteria 2 ***(A1)(C1)*** 75 % > 50 % (do not meet the requirement) ***(R1)(C1)*** **OR** 25 % < 50 % (meet the requirement) ***(R1)(C1)*** **Note:** Do not award (A1)(R0). Award the (R1) for a correct comparison of percentages for both cafeterias, which may be in words. The percentage values or fractions must be seen. It is possible to award (A0)(R1).
2.	**[1]**	125 (accept 125.5) ***(A1)***
3. (a)	**[1]**	42 (minutes) ***(A1)(C1)***
3. (b)	**[2]**	Stephen is correct. ***(A1)*** the minimum rehearsal time is greater than zero ***(R1)*** **OR** he rehearsed at least 20 minutes every day ***(R1)(C2)*** **Note:** Do not award (A1)(R0). Accept equivalent reasoning based on the box-and-whisker diagram.
3. (c)	**[3]**	0, 2, 3, 4, …, 15 ***(A1)(A1)(A1)(C3)*** **Note:** Award (A1)(A1) for each correct endpoint of the interval, (A1) for indication of integer values, except 1, between their endpoints.
4. (a)	**[1]**	9 (cm) ***(A1)(C1)***
4. (b)	**[1]**	40 (leaves) ***(A1)(C1)***
5. (a)(i)	**[1]**	$p = 10$ ***(A1)(C1)*** **Note:** Award (A1) for each correct value.
5. (a)(ii)	**[1]**	$q = 56$ ***(A1)(C1)*** **Note:** Award (A1) for each correct value.

5. (b)	**[2]**	 ***(A1)(A1)(C2)*** **Note:** Award (A1)(ft) for their 3 correctly plotted points; award (A1)(ft) for completing diagram with a smooth curve through their points. The second (A1)(ft) can follow through from incorrect points, provided the gradient of the curve is never negative. Award (C2) for a completely correct smooth curve that goes through the correct points.
5. (c)	**[2]**	a straight vertical line drawn at 35 (accept 35 ± 1) ***(M1)*** 26 (students) ***(A1)(C2)*** **Note:** Accept values between 25 and 27 inclusive.
6. (a)	**[1]**	16.5 ***(A1)(C1)***
6. (b)	**[3]**	 ***(A1)(A1)(A1)(ft)(C3)*** **Note:** Award (A1) for correct endpoints, (A1) for correct quartiles, (A1)(ft) for their median. Follow through from part (a)(ii), but only if median is between 16 and 18.5. If a horizontal line goes through the box, award at most (A1)(A1)(A0). Award at most (A0)(A1)(A1) if a ruler has not been used.

Section B

Question	Marks	Markscheme
7. (a)	**[2]**	valid approach ***(M1)*** *eg* $Q_3 - Q_1$, $Q_3 - 1.1$, $4.5 - a = 1.1$ $a = 3.4$ ***A1 N2***

7. (b)	**[1]**	$\frac{32}{5}$ (= 6.4) (km) ***A1 N1***
7. (c)	**[4]**	**METHOD 1 (standard deviation first)** valid approach ***(M1)*** *eg* standard deviation = $\sqrt{\text{variance}}$, $\sqrt{\frac{16}{9}}$ standard deviation = $\frac{4}{3}$ (km) ***(A1)*** valid approach to convert **their** standard deviation ***(M1)*** *eg* $\frac{4}{3}\times\frac{5}{8}$, $\sqrt{\frac{16}{9}} = \frac{8}{5}M$ $\frac{20}{24}$ (miles) (= $\frac{5}{6}$) ***A1 N3*** **Note:** If no working shown, award ***M1A1M0A0*** for the value $\frac{4}{3}$. If working shown, and candidate's final answer is $\frac{4}{3}$, award ***M1A1M0A0***. **METHOD 2 (variance first)** valid approach to convert variance ***(M1)*** *eg* $(\frac{5}{8})^2$, $\frac{64}{25}$, $\frac{16}{9}\times(\frac{5}{8})^2$ variance = $\frac{25}{36}$ ***(A1)*** valid approach ***(M1)*** *eg* standard deviation = $\sqrt{\text{variance}}$, $\sqrt{\frac{25}{36}}$, $\sqrt{\frac{16}{9}\times(\frac{5}{8})^2}$ $\frac{20}{24}$ (miles) (= $\frac{5}{6}$) ***A1 N3***
7. (d)	**[3]**	correct frequency for 22 minutes ***(A1)*** *eg* 20 adding **their** frequency (do not accept 22 + 400) ***(M1)*** *eg* 20 + 400 , 420 athletes *m* = 30 (minutes) ***A1 N3***

7. (e)	**[5]**	27 (minutes) ***(A1)*** correct working ***(A1)*** *eg* 130 athletes between 22 and 27 minutes, $P(22 < t < 27) = \frac{150 - 20}{600}$, $\frac{13}{60}$ evidence of conditional probability or reduced sample space ***(M1)*** *eg* $P(A \mid B)$, $P(t < 27 \mid 22 < t < 30)$, $\frac{P(22 < t < 27)}{P(22 < t < m)}$, $\frac{150}{400}$ correct working ***(A1)*** *eg* $\frac{\frac{130}{600}}{\frac{400}{600}}$, $\frac{150 - 20}{400}$ $\frac{130}{400}$ ($\frac{13}{40} = \frac{78000}{240000} = \frac{390}{1200} = 0.325$) ***A1 N5*** **Note:** If no other working is shown, award ***A0A0M1A0A0*** for answer of $\frac{150}{400}$. Award ***N0*** for answer of $\frac{3}{8}$ with no other working shown.
8. (a)	**[1]**	75 ***A1***
8. (b)	**[2]**	recognition that all entries add up to 120 ***(M1)*** $a = 120 - 6 - 13 - 26 - b$ OR $a = 75 - b$ ***A1***
8. (c)(i)	**[3]**	$\frac{6 \times 1 + 13 \times 2 + 26 \times 3 + (75 - b) \times 4 + b \times 5}{120} = 3.65$ ***(M1)(A1)*** **Note:** Award (M1) for attempt to substitute into mean formula, LHS expression is sufficient for the M mark. Award (A1) for correct substitutions in one variable OR in two variables, followed by evidence of solving simultaneously with $a + b = 75$. $b = 28$ ***A1***
8. (c)(ii)	**[2]**	120 − their part (c)(i) seen (e.g. 92 indicated on graph) ***(M1)*** 84 ***A1***

Please **do not** write on this page.

Answers written on this page
will not be marked.